

User Manual: Screen Reader ALO

Deliverable 5

Installation Guidelines

Part 1: Screen Reader All Installation

1. **Uninstall the previous version** if installed.
2. Click on the **Screen Reader Alo.exe** file.
3. Select the **location** on your computer where you want to install the software.
4. Click on the **Install** button to install the software.

Part 2: Bangla TTS Installation (Crucial for Bangla Speaking)

5. **Locate the Bangla TTS Installer.** This file is separate from the Screen Reader ALO installer.
6. Click on the **Bangla TTS Installer** file.
7. Follow the prompts to install the Text-to-Speech engine.
8. Once the TTS installation is complete, you may need to **reboot your computer**

Windows Application :

System

System-related hotkey that provides screen reader users with essential information for managing their device and environment more effectively.

No	Hotkey	Description
1	Alt + B	Announces the current battery status, including the battery level and charging status.
2	Alt + T	Reads aloud the current time, helping users keep track of the time without visually checking a clock.
3	Alt + D	States today's date, offering a quick way for users to know the date without needing to find a calendar.
4	Ctrl + T	Announces the title of the current window, allowing users to confirm they are in the desired application or document.
5	Shift	Interrupts any ongoing speech output, giving users control over the flow of auditory information and enhancing usability.
6	Alt + Insert	Application Restart
7	Alt + F	Exit the application forcefully
8	Alt + E	Exit Application with prompt window
9	Alt + Ctrl + Insert	Turn off/on the app without closing the app completely.

File Explorer

Support for File Explorer and basic navigation using Screen Reader functionalities.

No	Hotkey	Description
1	Up/Down/Left/Right	Enables navigation through individual elements, files, and folders, facilitating seamless exploration and selection.
2	Tab	Moves through accessible elements and groups, improving the efficiency of accessing various interface components.

3	F5	Announces the details of an element if the folder contains only one item, aiding in identifying contents without visual cues.
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Microsoft Word :

The Microsoft Word screen reader features are designed to enhance accessibility and usability for users with visual impairments. These features are divided into three main categories: General, Focus Mode, and Browse Mode.

General

No	Hotkey	Description
1	Ctrl + T	announces the current document's title, providing users with immediate context.
2	Ctrl+ Home	Moves the cursor to the document's beginning, allowing a quick return to the start.
3	Ctrl + Space	Toggles between Focus and Browse modes, adapting the interaction model to the task at hand.
4	Alt	Opens the Ribbon, Microsoft Word's toolbar that contains features and settings.
5	Tab	Enables traversal of the Ribbon Menu, voicing all available menu options to aid in selection.

Focus Mode on Microsoft Word

Focus Mode is tailored for editing and navigating within the text:

- Turn on Focus Mode using **CTRL + Space**

No	Hotkey	Description
1	Ctrl + Left Arrow	Moves the screen reader's cursor to the next word from the current position, and reads it aloud.
2	Ctrl + Right Arrow	Moves the screen reader's cursor to the previous word from the current position, reading it aloud.
3	Ctrl + Down Arrow	Advances the screen reader to the following paragraph, reading the first sentence or entire paragraph aloud, depending on the screen reader's settings.

4	Ctrl + Up Arrow	Moves the screen reader back to the previous paragraph, reading the first sentence or entire paragraph aloud, depending on the screen reader's settings.
5	Shift + Ctrl + Left Arrow	Selects and reads text one word at a time, moving left from the cursor's current position.
6	Shift + Ctrl + Right Arrow	Selects and reads text one word at a time, moving right from the cursor's current position.
7	Shift + Left Arrow	Selects and reads text one character at a time, moving left from the cursor's current position.
8	Shift + Right Arrow	Selects and reads text one character at a time, moving right from the cursor's current position.
9	Alt + Ctrl + A	Speak current table header properties (make sure the document has a table and the cursor is inside the table)
10	Up / Down arrow	Speak table properties like current row, column number & values (if inside a table)
11	Ctrl + Shift + X	Speak selected text style, size, font, etc. (make sure text is selected)
12	Space	When you type any word after giving a space, it will speak the word

Browse Mode

Browse Mode is optimized for reading and navigating the document structure:

- Turn on Browse Mode using **CTRL + Space**

No	Hotkey	Description
1	Esc	Allows the user to close the 'Restrict Editing' pop-up and return to the document.
2	W	Reads aloud the word immediately following the cursor's current position.
3	Shift + W	Read aloud the word immediately before the cursor's current position.
4	H	Navigates to and announces the next heading from the cursor's current position, aiding in document structure.

5	Shift + H	Navigates to and announces the previous heading from the cursor's current position, aiding in document structure.
6	C	Announces the character immediately following the cursor's current position.
7	Shift + C	Announces the character immediately before the cursor's current position.
8	Q	This action triggers the text-to-speech function to speak the word at the cursor position.
9	P	Upon activation, this hotkey instructs the text-to-speech engine to read aloud the entire paragraph immediately following the cursor's current location.
10	Shift + P	This function enables the text-to-speech system to vocalize the paragraph directly preceding the cursor's current point.

Microsoft Excel :

The Microsoft Excel screen reader features enhance accessibility for users with visual impairments through well-defined hotkeys. These shortcuts facilitate easy navigation, data editing, and interaction within Excel, making spreadsheet management more efficient for screen reader users.

No	Hotkey	Description
1	Tab	Allows for forward navigation through all types of menu and submenu items, enabling efficient exploration of options.
2	Shift + Tab	Allows for backward navigation through all types of menu and submenu items, facilitating easy backtracking and review.
3	Alt	Activates focus on Ribbon items, providing direct access to this toolbar for command selection.
4	Arrow Keys	Moves through Ribbon menu items when the Ribbon is focused, aiding in precise selection within this interface.
5	Up/Down/Left/Right	Enables navigation through spreadsheet cells, announcing the value and column name for each cell navigated.

6	F2 or Fn + F2	Initiates editing mode for the active cell, allowing for direct input or modification of cell content.
7	Ctrl + Shift + R	Announces the content of the formula bar, aiding in formula review or entry.
8	Ctrl + Right	When editing a cell, it moves and speaks the text word by word to the right, enhancing text navigation.
9	Ctrl + Left	When editing a cell, move the text to the left and speak it word by word, improving text review.
10	Control + Shift + S	announces the current sheet name, providing orientation within the spreadsheet.
11	Ctrl + Home	Moves the cursor to the first cell in the spreadsheet, allowing quick access to the starting point.
12	Ctrl + Page Up	Transition the view to the previous sheet in the workbook, facilitating easy navigation between multiple sheets.
13	Ctrl + Page Down	Advances the view to the next sheet in the workbook, streamlining navigation between sheets.
14	Left/Right arrow	Allows the user to hear the previous or next character in the active cell, enhancing accessibility and ease of data review. (press the 'home' key in the active cell, then navigate characters with left/right keys)
15	Delete	Upon deletion, the character removed from the cell is audibly announced, aiding in accurate data editing and ensuring correct information entry.
16	Ctrl+Delete	Upon deletion, the word removed from the cell is audibly announced, aiding in accurate data editing and ensuring correct information entry.
17	Home or End	Cell boundary announcement.

Microsoft PowerPoint :

The Microsoft PowerPoint screen reader features enhance accessibility for users with visual impairments through well-defined hotkeys.

No	Hotkey	Description
1	Alt	Activates focus on Ribbon items, allowing for direct access to commands and tools in the toolbar.
2	Shift + Tab	Moves backward through all menu and shape item types, enabling comprehensive navigation and review.
3	Tab	Navigates through shapes (including tables, charts, embedded objects, smart-art, pictures) or menus, facilitating detailed editing and customization.
4	Ctrl + M	Adds a new slide to the presentation, helping expand and organize content.
5	F5	Initiates the presentation from the beginning, which is helpful in rehearsing or delivering the presentation.
6	Shift + F5	The presentation begins on the current slide, which is convenient for focusing on specific sections.
7	Enter	Enables editing of the selected shape, providing a direct way to modify shapes while in editing mode.
8	Ctrl + Shift + C	Announces the current slide number, helping presenters keep track of their position in the presentation.
9	Ctrl + Shift + T	Announces the total number of slides in the presentation, offering a quick overview of presentation length.
10	Left/Right	Reads out the previous or next character, enhancing editing accuracy by providing auditory feedback on text input.
11	Escape	Pressing the Escape key allows the user to exit the slide thumbnail view and return to the presentation mode or the main interface.

12	N, Enter, Page Down, Right Arrow, Down Arrow, or Spacebar	Using any of these keys while in Presentation mode will advance the presentation to the next slide.
13	P, Page Up, Left Arrow, Up Arrow, or Backspace	These keys allow the presenter to go back to the previous slide during a presentation. This feature is essential for revisiting topics or correcting navigation mistakes without disrupting the flow of the presentation.
14	Control + D	By pressing Control + D, the user can create an exact copy of the selected slide.

Web Browser :

Chrome is a simple browser for navigating websites. Screen Reader supports some functionalities to speak aloud focused elements.

No	Hotkey	Description
1	Ctrl + Tab	Cycles through open menu items or tabs, enabling quick navigation without a mouse.
2	Ctrl + D	Save the current website to your favorites or bookmarks for easy access later.
3	Control + E	Initiates a Google search using the browser's default search engine.
4	Control + L	Moves cursor focus to the address bar, enabling direct entry of URLs or search queries.
5	Control + N	Creates a new tab, allowing for multiple pages to be viewed during the same browsing session.
6	Tab	Moves focus to the following link or accessible element on the webpage, facilitating forward navigation through interactive elements.
7	Shift + Tab	Moves focus to the previous link or accessible element, enabling backward navigation through interactive elements for thorough review.

8	Delete/Back	Speaks the deleted character while in editing
9	Left/Right	Speak the character it traverse over
10	Control + right	The cursor traverse over the word in the right direction (left to right) and speaks the word, in front of which is the cursor,
11	Control + left	The cursor traverse over the word in the left direction (right to left) and speaks the word, in front of which is the cursor,

Feature

No	Hotkey	Description
1	Up + Control+ Alt	Moves to and reads aloud the next element on the web page, enhancing forward navigation.
2	Down + Control +Alt	Moves to and reads aloud the previous element, aiding in backward navigation through content.
3	H + Control + Alt	Advances to the next header element and reads it aloud, which is helpful in navigating through sections of content.
4	H + Shift + Alt +Control	Moves to and reads aloud the previous header, facilitating easy review of content structure.
5	T + Control + Alt	Navigates to the following table and reads its content, assisting in understanding data presented in table format.
6	T + Control + Shift + Alt	Moves to and reads aloud the contents of the previous table, enabling a comprehensive review of tabular data.

YouTube :

YouTube key bindings are automatically activated when you navigate to YouTube in your browser. The screen reader detects YouTube and provides enhanced navigation and video control features.

No	Hotkey	Description
1	Tab	Navigate to the front when the video is found.
2	Shift + Tab	Navigate to the previous Elements/ video in the list and announce its details.

3	Search bar	work as the Search bar as usual & after pressing space, speak the last word
4	M	Mute/Unmute

Facebook:

Facebook key bindings are automatically activated when you navigate to Facebook in your browser. The screen reader provides enhanced navigation for posts, comments, and Facebook-specific content.

Pressing J or K moves the focus between posts. This action may not automatically read the content aloud, but it correctly shifts the focus to the selected post. After the focus moves, you can press Ctrl + Alt + Up/Down Arrow to read and navigate through the content. This is the typical way web browser screen readers operate.

To achieve the best user experience, it is recommended to combine different keyboard shortcuts rather than relying on a single command.

You can also use Shift and Shift + Tab for basic navigation and traversal between elements. Many shortcuts here are here to AID IN NAVIGATION!

N.B.: Enable Facebook Web Single Shortcuts/Accessibility from Settings Via Facebook Settings (Recommended)

1. Go to Facebook.com in Chrome
2. Click your profile picture (top right)
3. Click "Display and Accessibility."
4. In the left sidebar, click "Keyboard."
5. Turn on, Use single-character keyboard shortcuts

No	Hotkey	Description
1	J	Its purpose is to shift the focus to the next post. Navigate to the next post in the News then use Ctrl+ Alt + down to read details and if you want to read backwards use Ctrl + Alt + up

2	K	Its main purpose is to shift the focus to the previous post. Navigate to the previous post in the News, use Ctrl+ Alt + down to read details and if you want to read backwards use Ctrl + Alt + up
3	L	Its purpose is to shift the focus to the reaction panel. Pop up reaction panel, 'Like' reaction is pre-selected, so reader speaks it, you can use Left/Right arrow keys to traverse between Reactions and the reader will speak it.
4	C	Its purpose is to pop up a window so that users can comment. Use Tab or Shift+Tab or Ctrl+Alt+up/down to get it SPEAK the content
5	S	Its purpose is to open the Share dialog for a selected post. Use Tab or Shift+Tab or Ctrl+Alt+up/down to get it SPEAK the content
6	O	Its purpose is to open the post. Use Tab or Shift+Tab or Ctrl+Alt+up/down to get it SPEAK the content
7	P	It's purpose is to Open the status composer to create a new post Use Tab or Shift+Tab or Ctrl+Alt+up/down to get it SPEAK the content
8	/	Focuses on the Facebook search bar, and speaks what is focused. Use Tab or Shift+Tab or Ctrl+Alt+up/down to get it to SPEAK the content since the focus is moved to the Search Bar.
9	Q	Q is not supported in all types of facebook accounts/ui. It is used to search Messenger Contacts: When in the main News Feed, pressing Q opens the search box to send messages to friends/contacts.

NOTE: Facebook-specific shortcuts only work when Facebook is the active tab. The screen reader automatically detects Facebook and enables these features. Enhanced announcements include post author, timestamp, reaction counts, comment counts, and share counts.

Gmail:

Gmail key bindings are automatically activated when you navigate to Gmail in your browser. The screen reader provides comprehensive support for Gmail's native keyboard shortcuts, making email management efficient and accessible.

****** for Gmail, need to turn on Gmail hotkeys from settings**

***** Must enable Gmail shortcuts on**

-> go to settings->see all settings-> Keyboard shortcuts on -> save Changes

Navigation & Views

No	Hotkey	Description
1	down arrow	Move to the next (older) conversation in the list and announce the sender, subject, snippet, and metadata.
2	up arrow	Move to the previous (newer) conversation in the list and announce details
3	Enter	Open the selected conversation and read the email content
4	U	Return to inbox view from conversation view
5	N	Move to the following message within an open conversation
6	P	Move to the previous message within an open conversation
7	g + i	Navigate to the inbox view
8	g + a	View all mail
9	g + s	View starred messages
10	g + t	View sent mail
11	g + d	View draft mail
12	g + b	View snoozed mail

Compose, Reply, and Formatting

No	Hotkey	Description
13	C	Open a compose window for a new email (Compose mail announce0
14	D	Open a compose window in a new browser tab
15	Ctrl + Enter	Send the composed email (reads status banner)
16	R	Reply to the sender when you open an email
17	F	Forward the email

Message Actions & Status

No	Hotkey	Description
18	g+l ->press spam ->press Enter	When press g+l ..its open label of gmail.. then need to type Spam..after after press enter will announce Spam mail.

Selection & Utility

No	Hotkey	Description
19	?	Open Gmail keyboard shortcuts help.
20	/	Focus the search box.
21	Space (in search bar)	When typing in the Gmail search bar, pressing Space announces the completed word.

IMPORTANT NOTES FOR GMAIL

- Sequence shortcuts (g prefix and * prefix) require pressing the second key within 800ms of the prefix.
- The screen reader announces "G prefix" or "Star prefix" when you press the prefix key.

- Navigation shortcuts (J, K, O, N, P) wait 300ms to read conversation details after navigation.
- Archive, delete, and spam actions wait 200ms to read Gmail's confirmation banner.
- Gmail shortcuts only work when Gmail is the active tab
- Enhanced announcements include sender, subject, snippet, timestamp, and email content.

Zoom:

General Controls & View

In case of Zoom chat, you can use Tab, Shift + Tab, Up/Down Arrow, and mind map like how visually impaired testers do to get the maximum user experience.

1. Even focus is lost, try to gain focus, use Tab, Shift+Tab to navigate and gain focus, let the zoom screen reader start and settle Hold Alt, then press other corresponding keys
2. In case of reactions, Hold Alt+Shift, then press other corresponding keys
3. while testing, don't be fast, zoom via walk is slow due to multi process ranging from 3 to 7 during phases, and Inter process communication
4. Try to gain focus, and use Tab or Shift+Tab to activate Zoom interaction or re focus windows if nothing is spoken, wait for few seconds to get it activated in case of any delay or lag

No	Hotkey	Description
1.	Alt + A	Speaks Audio now muted/unmuted
2.	Alt + V	Speaks Video now started/stopped
3.	Alt + H	Speaks about Chat Panel, also speak if chat panel is closed
4.	Alt + U	Speaks about Participants Panel, also speak if chat panel is closed

5	Alt + S	Speaks about Screen sharing or the button focused
6	Alt + R	Zoom default speaks out about recording on or off
7	Alt + C	Speaks about Cloud Recording (host)
8	Ctrl + N	Start New Chat, Speak about it (it should be used when the chat window has been opened using Alt+H)
9	Alt + Q	Leave the current meeting
10	Ctrl + 2	Speaks out the active speaker
11	Alt + F1	Speaks Switch to speaker view mode
12	Alt + F2	Speaks Switch to gallery view mode
13	PageUp	View previous page of participants in gallery, speaks about it
14	PageDown	View next page of participants in gallery, speaks about it
15	Up Arrow	- Speaks the message above in chat - speaks the menu tab during navigation
16	Down Arrow	- Speaks the message above in chat - speaks the menu tab during navigation
17	Alt+Shift+E	- Zoom Focus on the edit message in chat - speaks about it
18	Alt + Shift + I	Speak invite link copied to clipboard
19	Alt + Y	Speaks Hand raised / Hand lowered
20	Tab	It can traverse between intricate chat window properly and speaks focus item [FORWARD]
21	Shift + Tab	It can traverse between intricate chat window properly and speaks focus item [BACKWARD]

WhatsApp:

Chat Management

No	Hotkey	Description
1	Ctrl + N	Focus moves to new Chat
2	Ctrl + F	Announces search field
3	Up Down arrow	Announce the Chat details
4	Enter	Enter the chat
5	Ctrl + Shift + F	Reads search context
6	Ctrl + P	Open Profile
7	Ctrl + Shift + M	Announces chat muted

UI, Text Navigation, and Deletion

No	Hotkey	Description
8	Tab	Announces focused element
9	Shift + Tab	Announces focused element

10	Up Arrow	Reads previous message/line/element
11	Down Arrow	Reads next message/line/element
12	Alt + S	Open Settings